AI-Powered E-Commerce Sales Analytics & Forecasting

Portfolio project report | Synthetic dataset | October 2026

Executive Summary

This educational project generates 10,933 synthetic orders across 2023–2025, analyzes revenue and profitability, and predicts daily sales. All values are simulated and must not be represented as real business performance.

Workflow

1. Generate transaction records with dates, customers, products, regions, prices, discounts, sales and profit. 2. Aggregate daily revenue. 3. Query key metrics using MySQL. 4. Visualize trends in Streamlit or build a Power BI report from supplied measures. 5. Fit a Random Forest forecasting model and evaluate on a later chronological period.

Evaluation

Training period: through June 30, 2025. Holdout: July 1–December 31, 2025. Mean absolute error: INR 19,090.47. Root mean squared error: INR 24,023.97. MAPE: 84.28%. Holdout uses actual past sales as lag features (one-step ahead); the next 30 days are recursively forecast and uncertainty increases with horizon.

Limitations

Synthetic data can be easier to model than real demand. No external factors such as promotions, returns, inventory or holidays are included. Forecasts are illustrative, not operational guidance. A seasonal naive baseline and rolling-origin validation should be added before real-world use.

How to Demonstrate

Open the Streamlit dashboard, explain each KPI, show SQL queries, discuss the chronological split and error metrics, then recreate visuals in Power BI Desktop using the DAX guide. Explain that a native PBIX file is not included.